MAJOR PROJECT REPORT ON

Audio Recording App

"**Easy-Record**"

UNDERTAKEN AT

SANT SINGAJI INSTITUTE OF SCIENCE AND MANAGEMENT

BY

Naman Saran

SUBMITTED TO

VIKRAM UNIVERSITY UJJAIN (M.P.)

Partial fulfilment of the Requirements for the Award of the Degree Of

Bachelor of Computer Application

(B.C.A.) March - April 2024

Guide Name

Mr. Yogendra Rajput

Sant Singaji Institute of Science and Management Sandalpur [Dewas]

Certificate

Reference No: ……. Date: ……

Project Completion Certificate

This is to certify that **Mr. Naman Saran** student of BCA **(Final Year)** of **Sant Singaji Institute of Science and Management**, has successfully completed the project work entitled **"Easy-Record"** under the guidance of **Mr. Yogendra Rajput**is a bona fide piece of work carried out at **Sant Singaji Institute of Science and Management Sandalpur.**

The project entitled **"Easy-Record"** developed by **Naman Saran** in the **SSISM** and he has put at least 200 hours of laboratory work during the tenure of the projectwith the guide to complete this project. All the **prescribed certificates** are attached after the completion of all the formalities of the project work as pre- schedule, including internal examination.

Place **Signature of Principal**

Date **Signature of HOD**

Certificate of Attendance

Reference No: ……. Date: ……

This is to certify that **Mr. Naman Saran,** Student of BCA **(Final Year)** of **Sant Singaji Institute of Science and Management**, has successfully completed the project work entitled **"Easy-Record"** and he has put at least 200 hours of laboratory work during the stipulated period of the project at **Sant Singaji Institute of Science and Management, Sandalpur.**

Place **Signature of Guide**

Date **Signature of Principal**

Declaration

I'm **Naman Saran** of **Sant Singaji Institute of Science & Management** declarethat the project report submitted by me under the guidance of **Yogendra Rajput** is a bona fide work for the partial fulfilment of the requirement of the **BCA Final Year project work**. I have incorporated all the suggestions provided by my guide time to time.

I further declare that to the best of my knowledge this project contains my original work & does not contain any part of any work which has been submittedfor the award of any degree either in this university or in any other university/Deemed university/Institute etc. Without proper citation and I shall befully responsible for any plagiarism found at any stage.

Name & Signature of the Guide **Name & Signature of the Student**

Project Approval Certificate

Reference No: ……. Date: ……

This to certify that **Mr. Naman Saran, a** student of **BCA (Final Year)** of **Sant Singaji Institute of Science & Management**, has successfully completed the project entitled **"Easy-Record"** under my guidance. I have regularly assessed the progress of the work and suggested the correction whenever required. The student has incorporated all the suggestions provided by me in this project. This project is bona fide piece of work of the standard of BCA project work carried out by the student under my supervision. Internal examination has been completed in my presence and student's performance was satisfactory and hence this project is approved for the submission and valuation thereof.

Signature of Guide

Place: **Signature of Principal**

Date: **Seal of the Institute**

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Introduction

✓ Project Overview

In **“Easy-Record”**, we are trying to implement an Audio Recording Application. **“Easy-Record”** is a Mobile Application which allows users to Record Audio in Raw data in PCM.

My-Shop is a mobile app through which you can record audio anytime and it’s very user-friendly. Being a mobile Application, it allows users to access the facility anywhere anytime.

In today's time, All the audio recording app is only providing audio record feature but if user want to save details about who is doing recording with whom. So, for that we have develop this application in that you can save user details with images.

You can record audio in multiple formats with this application. We will provide on dropdown to select different type of format to do recording. And we also give option for sampling rate so user can customize the recording settings based upon their requirements of audio. **“Easy-Record”** has one phases user phase.

User Phase

In User phase user can login our self by filling mobile number. After filling we will provide them 6-digit OTP. After successfully verification of OTP user can login to the app.

In the home screen user can see list of recorded audios. Or if he wants to create new recording then he can click on Create Recording button so he can create recording.

When user fill his details and select images, he can go to record audio and after recording audio he can play recorded audio in the Preview screen. After doing success full recording user can navigate to home screen and he can see the details about recorded audio.

1.2 Problems and solutions of the project

1.2.1 Existing System

In the present scenario, Audio Recording app are only providing the record audio feature, we can only record the audio and save that audio or we can share that audio but if user want to record audio with user details and images for that there is no such option to record the audio.

1.2.2 Proposed System

In our proposed system we are trying to solve all these problems which are faced in existing system. In **"Easy-Record"** we are trying to solve the problemsof the people who want to record audio with user details and images. Our aim is to create a app whereall the people can record audio easily with no complexity.

Feasibility Report

Preliminary investigation examines project feasibility, the likelihood the system will be useful to the organization. The main objective of the feasibility study is to test the Technical, Operational and Economical feasibility for "**Easy-Record**" adding new modules and debugging old running system. All system is feasible if they are unlimited resources and infinite time. There are aspects in the feasibility study portion of the preliminary investigation:

- ✓ Technical Feasibility
- ✓ Operation Feasibility
- ✓ Economic Feasibility

2.1. Technical Feasibility

The technical issue usually raised during the feasibility stage of the investigation includes the following:

- Does the necessary technology exist to do what is suggested?
- Can the system be upgraded if developed?
- Are there technical guarantees of accuracy, reliability, ease of access anddata security?

Earlier no system existed to cater to the needs of 'Secure Information Implementation System'. The current system developed is technically feasible. It is a web-based user interface for Online Exam audit workflow at NIC- CSD. Thus, it provides an easy access to the users. The database's purpose is to create, establish and maintain a workflow among various entities in order to facilitate all concerned users in their various capacities or roles. Permission to the users would be granted based on the roles specified.

2.2. Operational Feasibility

Proposed projects are beneficial only if they can be turned out into information system. That will meet the organization's operating requirements. Operational feasibility aspects of the project are to be taken as an important part of the project implementation. Some of the important issues raised are to test the operational feasibility of a project includes the following: -

- Is there sufficient support for **"Easy-Record"** from the users?
- Will the system be used and work properly if it is being developed and implemented?

2.3. Economic Feasibility

A system can be developed technically and that will be used if installed must still be a good investment for **"Easy-Record"** for the organization.

System Analysis

Software Requirement Specification Overview

We aim to become pioneers in the Audio Recording App with the first step of "**Easy-Record"** by fulfilling the requirements of the users. All of these elements will drive us towards success and show us as one company that can perform and give value for Digitalization. This Service will solve many people's problems because we have facilitated our users with many facilities so that our customers didn't have to go here and there and find some random App to do the same. All these things waste the valuable time of people in this high-tech busy world so with the help of our "**Easy-Record"** Audio Recording App people will save their valuable time and money.

3.1 Home module

Through the home module, the user can access our various functionalities.

1. Home Module

Through the home module, the user can access our various functionalities. On our login page, the user will login into "**Easy-Record"** after login user will see home page in our home page user can see history of recorded audios and he can record new audio through Crete Recording. This module consists of various screens according to user requirements which enable a user to get a better experience with a systematic design

User Functionalities:

- Users can log in or Sign In.
- Users can see recorded audios.
- User can create new audio

2. User details and Image select Module

Through the home module, user can access our various functionalities. On our home page, the user will interact with "**Easy-Record" in** various sections like Create Recording and Image select, user can fill the user details and he can select images for recording. This module consists of various screens according to user requirements which enable a user to get a better experience with a systematic design

User details and Image select Functionalities:

- User can fill user's details.
- User can select and take Images.
- User can preview the selected Images.

3.2 Audio Record Module

In our "**Easy-Record"** there is an Audio Record module page that will allow users to record the audio. In Record audio user can record after recording is done user can play the recorded audio and if user want to re-record user can record audio again. Once recording is done user can go to home screen and see the detail of the recorded audio with images and user details.

Functionalities:

- Users can record the audio.
- Users can re-record the audio.
- User can play the recorded audio.
- Users can see list of recorded audios.
- User can see details of the recorded audios.

System Requirement Analysis

4.1 Hardware Requirements

- Android mobile device is needed.
- 3 GB RAM and above.
- 32 ROM and above.

4.2 Software Requirements

- Android operating system is needed.
- Android Version is above 9
- Internet Connection.

System Design

5.1 Module Design

Software design sits at the technical kernel of the software engineering process and is applied regardless of the development paradigm and area of application. Design is the first step in the development phase for **"Easy-Record"** Audio Recording app or any engineered product or system. The designer's goal is to produce a model or representation of an entity that will later be built. Beginning, once system requirement has been specified and analyzed, system design is the first of the three technical activities -design, code and test that is required to build and verify software.

The importance can be stated with a single word "Quality". Design is the place where quality is fostered in software development. Design provides us with representations of software that can assess for Audio Recording platform **"Easy-Record"** quality. Design is the only way that we can accurately translate a user's view into a finished software product or system. Software design serves as a foundation for all the software engineering steps that follow. Without a strong design, we risk building an unstable system – one that will be difficult to test, onewhose quality cannot be assessed until the last stage.

During design, progressive refinement of data structure, program structure, and procedural details are developed reviewed and documented. System design can be viewed from either technical or project management perspective. From the technical point of view, design is comprised of four activities – architectural design, data structure design, interface design and procedural design

5.2 Data Flow Diagrams

A data flow diagram is graphical tool used to describe and analyze movement of data through a system. These are the central tool and the basis from which the other components are developed. The transformation of data from input to output, through processed, may be described logically and independently of physical components associated with the system. These are known as the logical data flow diagrams. The physical data flow diagrams show the actual implementsand movement of data between people, departments and workstations. A full description of a system actually consists of a set of data flow diagrams. Using twofamiliar notations Yourdon, Gane and Sarson notation develops the data flow diagrams. Each component in a DFD is labeled with a descriptive name. Process is further identified with a number that will be used for **"Easy-Record"** audio recording app identification purpose. The development of DFD'S is done several levels. Each process in lower-level diagrams can be broken down into a more detailed DFD in the next level. The top-level diagram is often called context diagram. It consists a single process bit, which plays vital role in studying the current system. The process in the context level diagram is exploded into other process at the first level DFD.

DFD Symbols

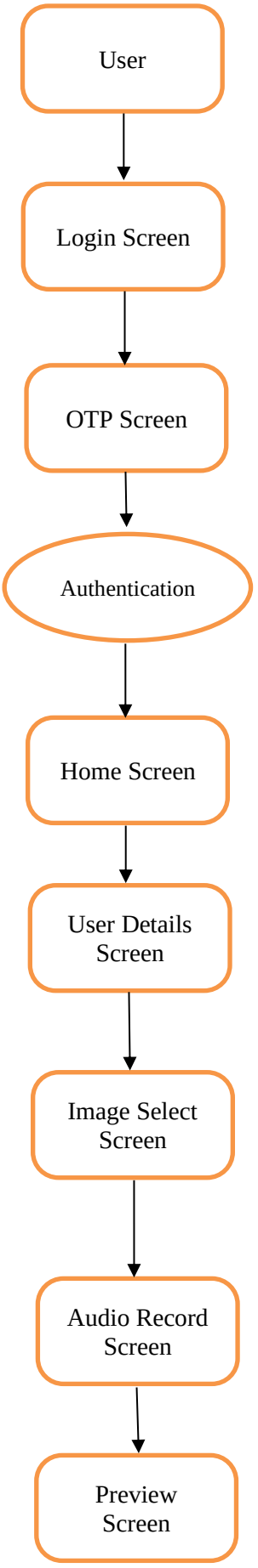
In the DFD, there are four symbols

1. A square defines a source (originator) or destination of system data.
2. An arrow identifies data flow. It is the pipeline through which the informationflows.
3. A circle or a bubble represents a process that transforms incoming data flowinto outgoing data flows.

Data Flow

1. A Data Flow has only one direction of flow between symbols. It may flow inboth directions between a process and a data store to show a read before an update. The latter is usually indicated however by two separate arrows since these happen at different type.
2. A join in DFD means that exactly the same data comes from any of two or more different processes data store or sink to a common location.
3. A data flow cannot go directly back to the same process it leads. There must be at least one other process that handles the data flow produce some other data flow returns the original data into the beginning process.
4. A data flow to a data store means update (delete or change).
5. A data flow from a data store means retrieves or use.

A data flow has a noun phrase label more than one data flow noun phrase can appear on a single arrow as long as all of the flows on the same arrow move together as one package.


User
Login Screen
OTP Screen
Authentication
Home Screen
User Details Screen
Image Select Screen
Audio Record Screen
Preview Screen

Firebase Database

Firebase Storage is a cloud-based storage service by Google, offering secure and scalable storage for user-generated content such as images, audios, videos and files. With features like powerful security rules, easy integration with Firebase Authentication, and SDKs for multiple platforms, it provides developers with a reliable solution for storing and serving content to their applications.

Firebase Storage

In Firebase Storage, files are structured according to their paths or filenames, providing a systematic organization method. This organization simplifies the retrieval process and ensures efficient management of stored files.

Multi-Version Concurrency Control (MVCC)

Multi-Version Concurrency Control (MVCC) is an advanced technique for improving database performance in a multi-user environment

System Testing and Implementation

6.1. Introduction

Software testing is a critical element of software quality assurance and represents the ultimate review of specification, design and coding. In fact, testing is the one step in the software engineering process that could be viewedas destructive rather than constructive.

A strategy for **"Easy-Record"** software testing integrates software test case design methods into a well-planned series of steps that result in the successful construction of software. Testing is the set of activities that can be planned in advance and conducted systematically. The underlying motivation of program testing is to affirm software quality with methods that can economically and effectively apply to both strategic to both large and small- scale systems.

6.2. Software Testing

The software engineering process can be viewed as a spiral. Initially system engineering defines the role of software and leads to software requirement analysis where the information domain, functions, behavior, performance, constraints and validation criteria for **"Easy-Record"** software are established. Moving inward along the spiral, we come to design and finally to coding. To develop computer software, we spiral in along streamlines that decrease the level of abstraction on each turn.

A strategy for **"Easy-Record"** software testing may also be viewed in the context of the spiral. Unit testing begins at the vertex of the spiral and concentrates on each unit of the software as implemented in source code.
Testing progress by moving outward along the spiral to integration testing, where the focus is on the design and the construction of the software

architecture. Talking another turn on outward on the spiral we encounter validation testing where requirements established as part of software requirements analysis are validated against the software that has been constructed. Finally, we arrive at system testing, where the software and othersystem elements are tested as a whole.

6.3. Unit Testing

Unit testing focuses verification effort on the smallest unit of software design, the module. The unit testing, we have is white box oriented and some modules the steps are conducted in parallel.

1. White Box Testing

This type of testing ensures that

- All independent paths have been exercised at least once.
- All logical decisions have been exercised on their true and false side.
- All loops are executed at their boundaries and within their operational bounds.

2. Basic Path Testing

Established technique of flow graph with Cyclometric complexity was used to derive test cases for **"Easy-Record"** all the functions. The main steps in deriving test cases were:
Use the design of the code and draw correspondent flow graph.

Determine the Cyclometric complexity of resultant flow graph, using formula: V(G)=E-N+2 or

V(G)=P+1 or

V(G)=Number of Regions
Where V(G) is Cyclomatic complexity,E is
the number of edges,
N is the number of flow graph nodes,

P is the number of predicate nodes.

Determine the basis of set of linearly independent paths.

3. Conditional Testing

In this part of the testing each of the conditions were tested to both true and false aspects. And all the resulting paths were tested. So that each path that may be generate on particular condition is traced to uncover any possible errors.

4. Data Flow Testing

This type of testing selects the path of the program according to the location of definition and use of variables. This kind of testing was used only when some local variable was declared. The definition-use chain method was used in this type of testing. These were particularly useful in nested statements.

5. Loop Testing

In this type of testing all the loops are tested to all the limits possible. The following exercise was adopted for **"Easy-Record"** all loops:

- All the loops were tested at their limits, just above them and just below them.
- All the loops were skipped at least once.
- Unstructured loops were resolved into nested loops or concatenated loops andtested as above.
- Each unit has been separately tested by the development team itself and all theinput have been valid

Output Screen and Code

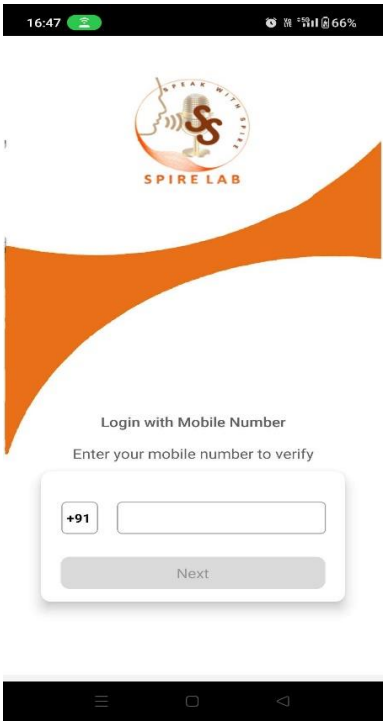


Fig. Login Screen

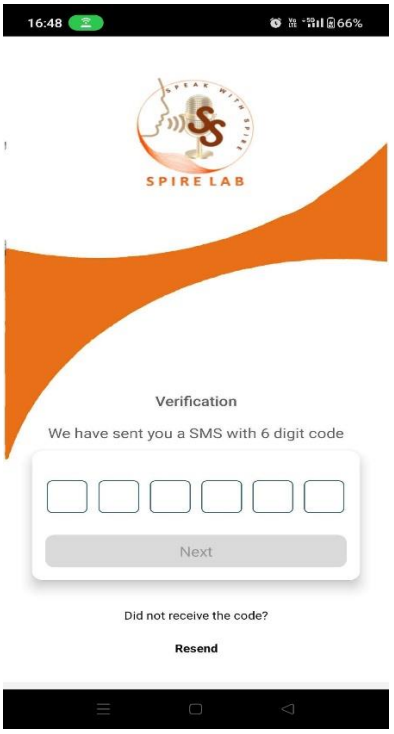


Fig. OTP Verification Screen

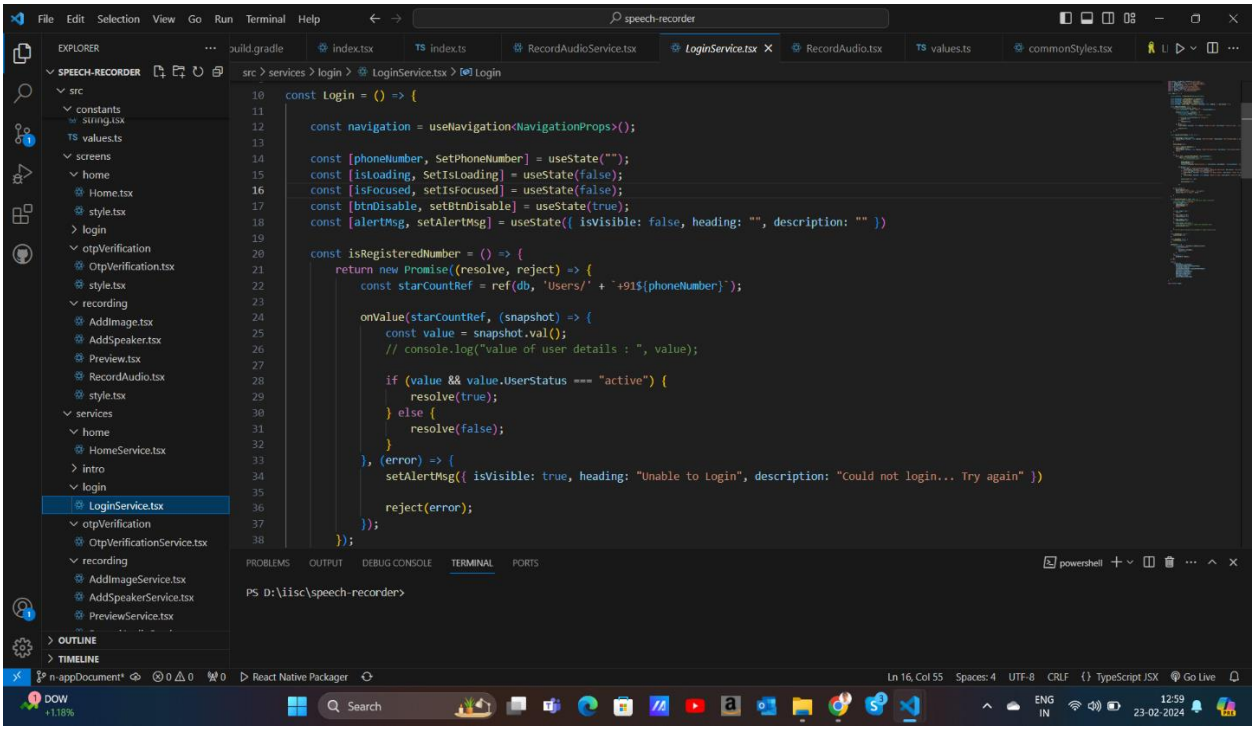


Fig. Login Screen Code

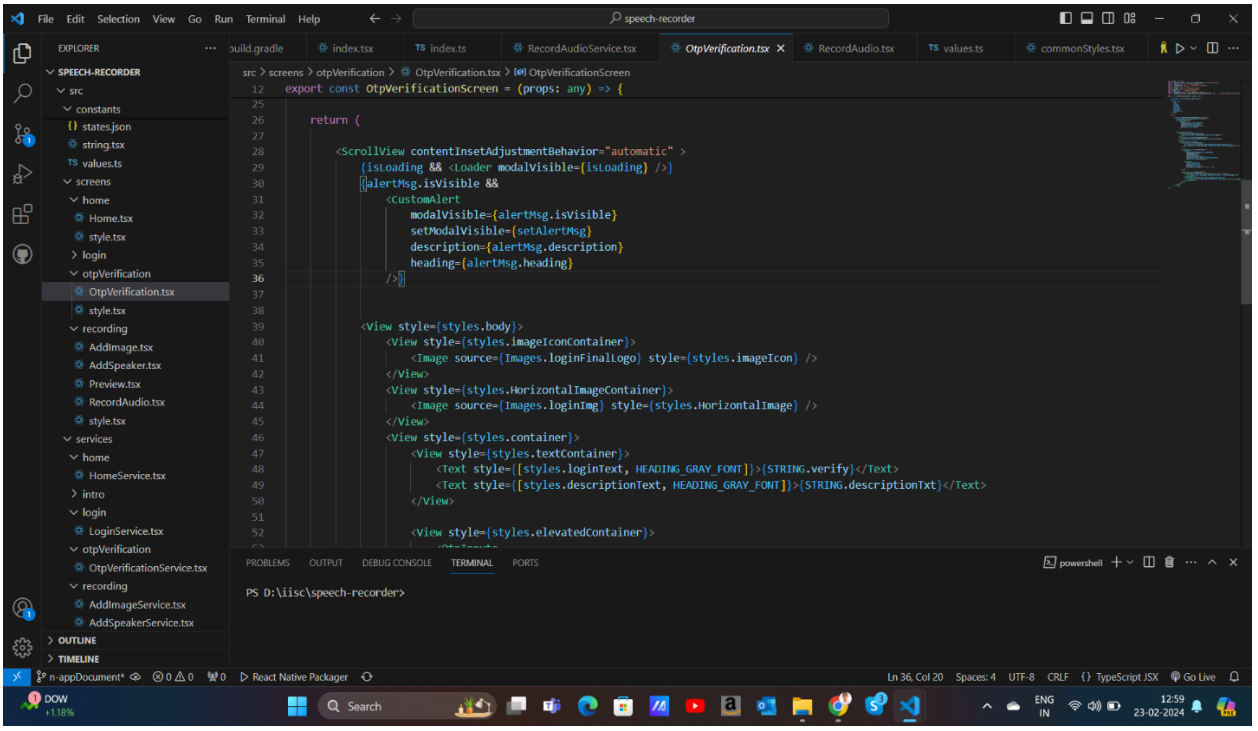


Fig. OTP Screen Code

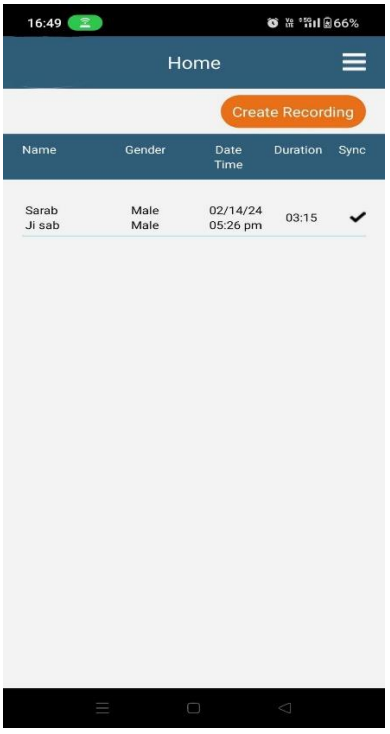


Fig. Home Screen

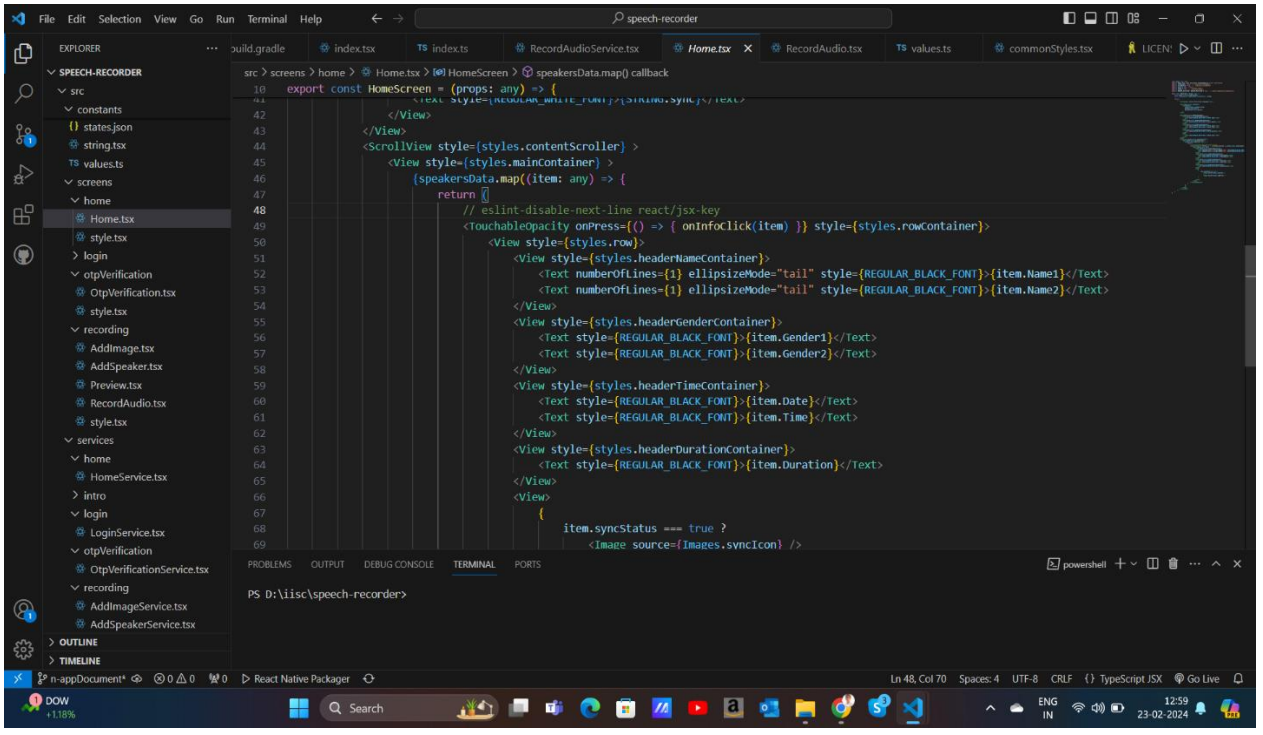


Fig. Home Screen code

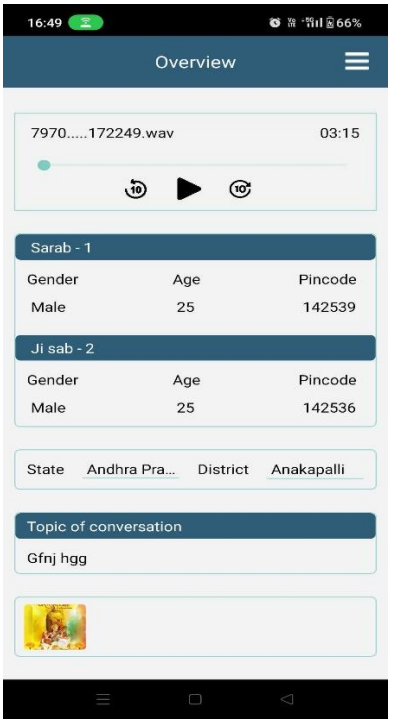


Fig. Overview Screen

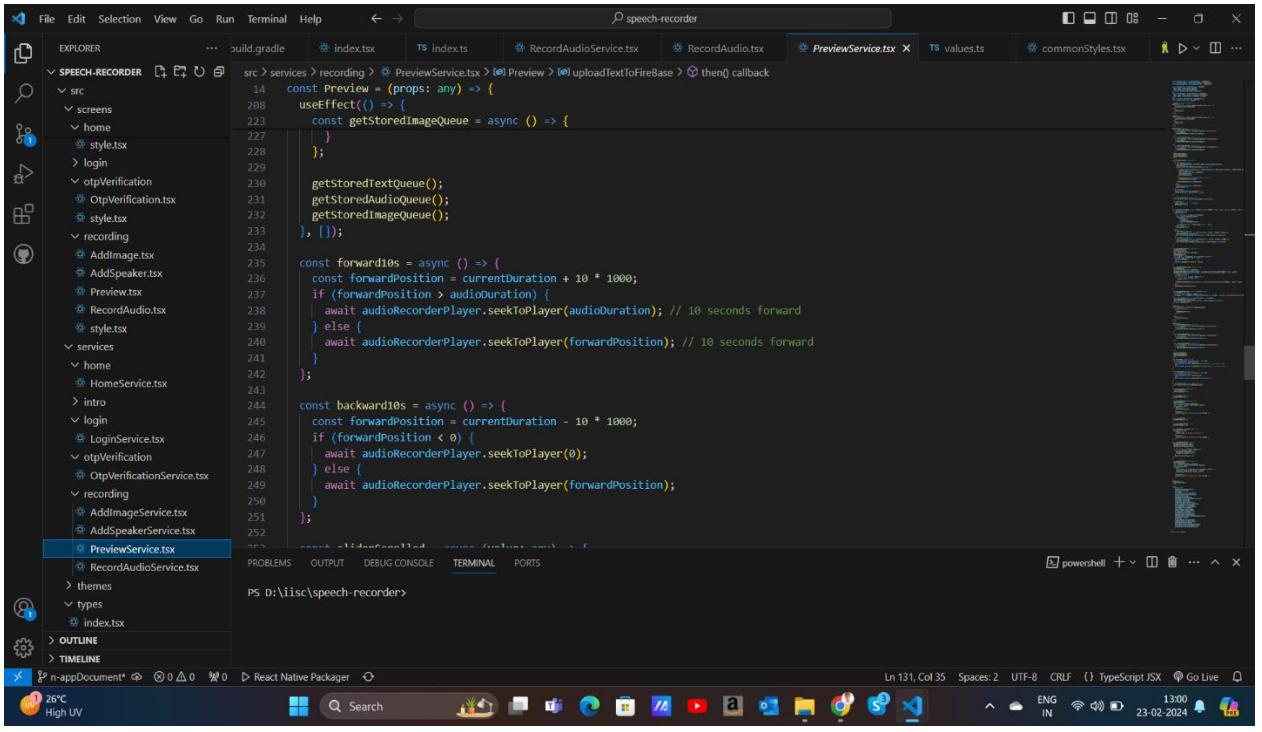


Fig. Overview Screen Code

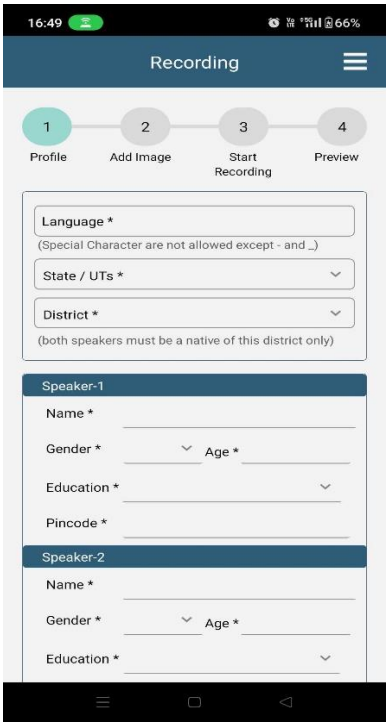


Fig. User details screen

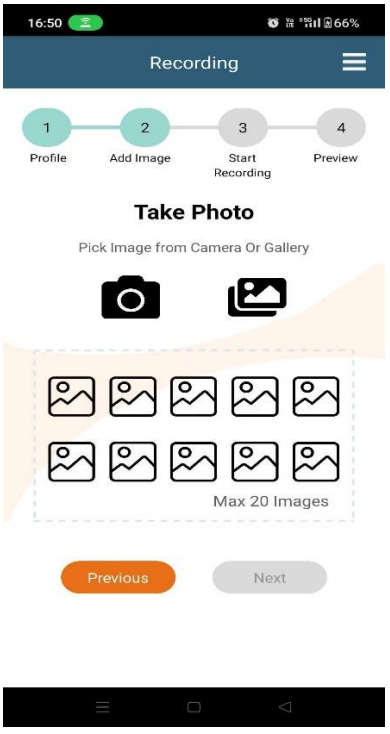


Fig. Select Image Screen

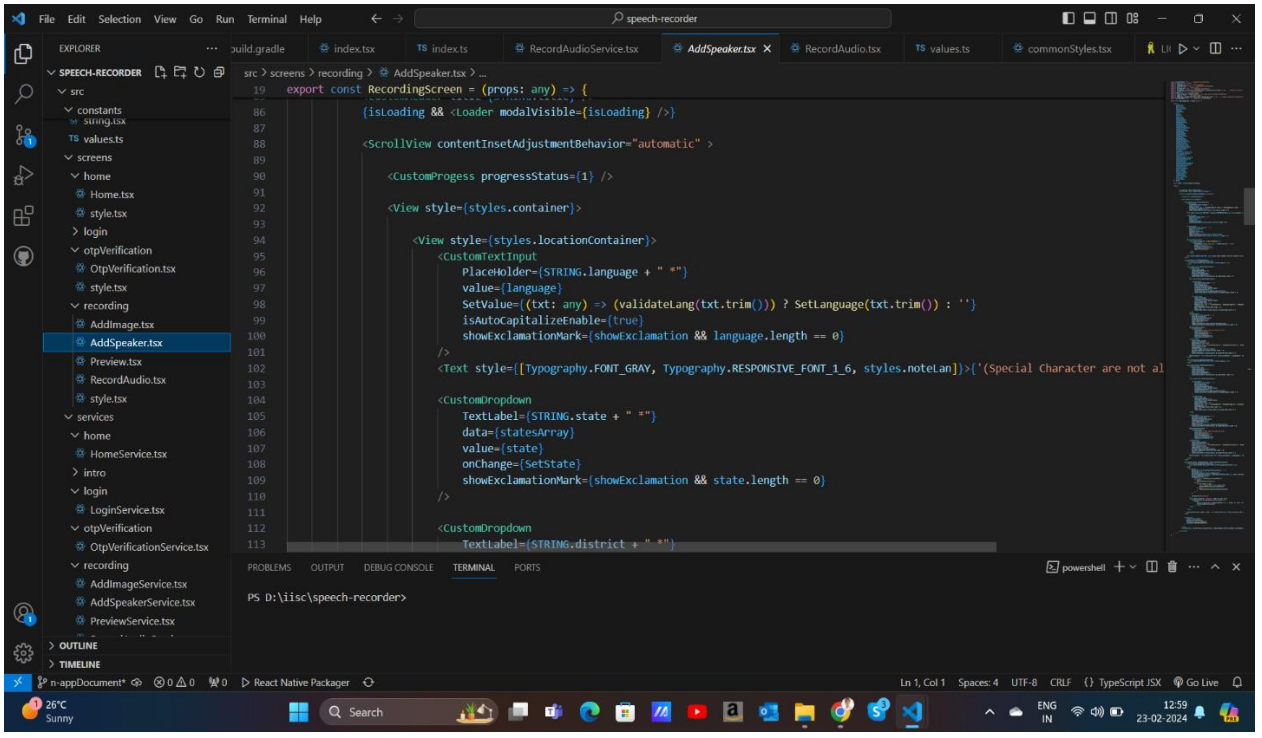


Fig. Add User Details Screen Code

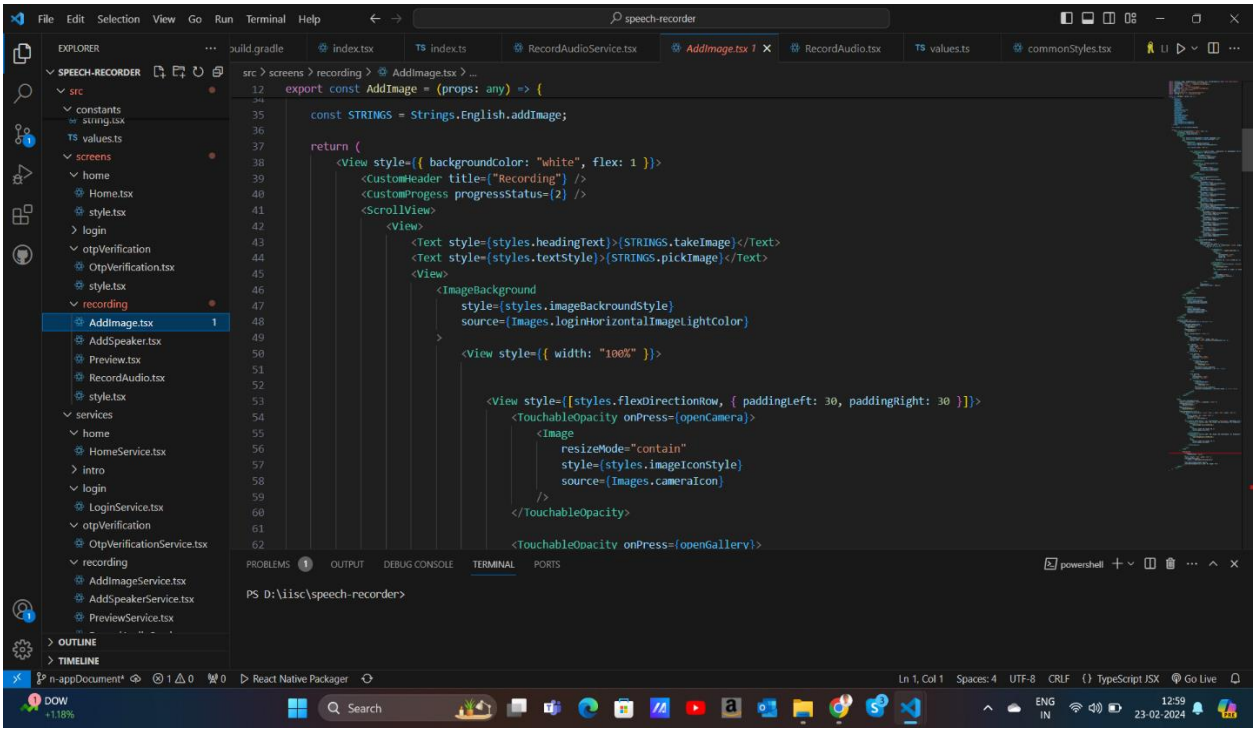


Fig. Add Image Screen Code

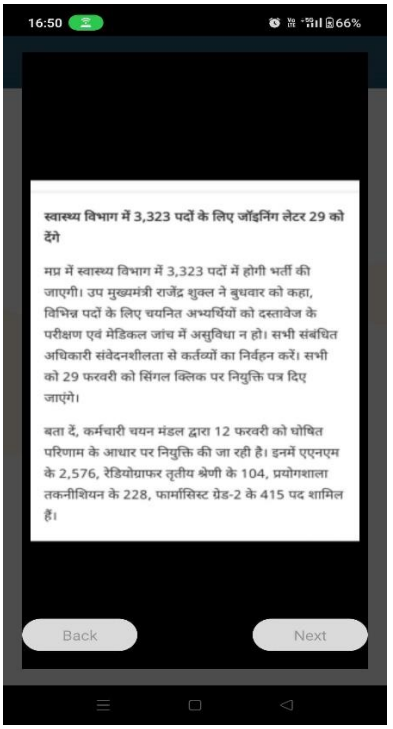


Fig. Preview Selected Image

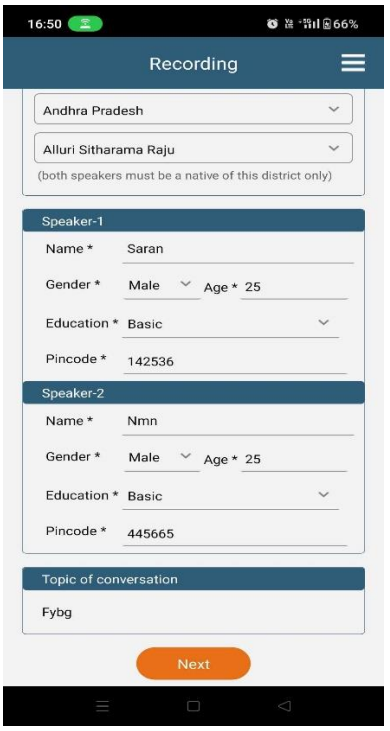


Fig. Filled User Details Screen

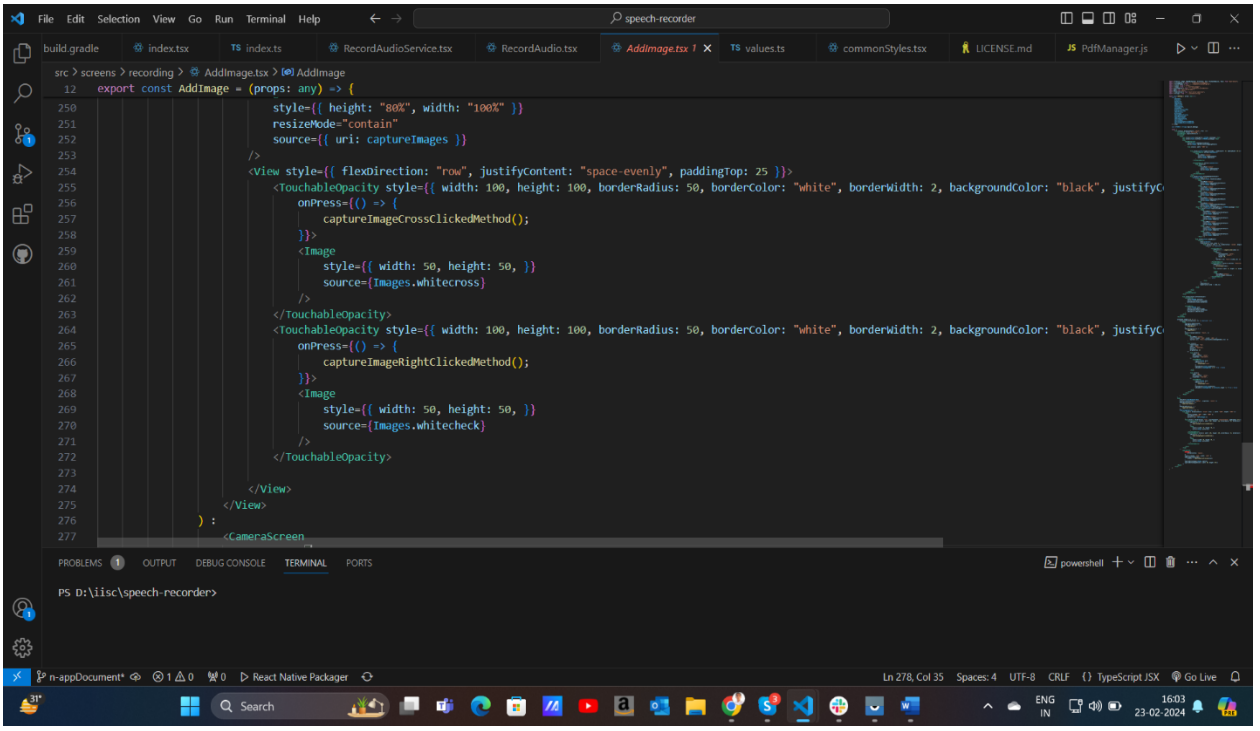


Fig. Select Image screen code

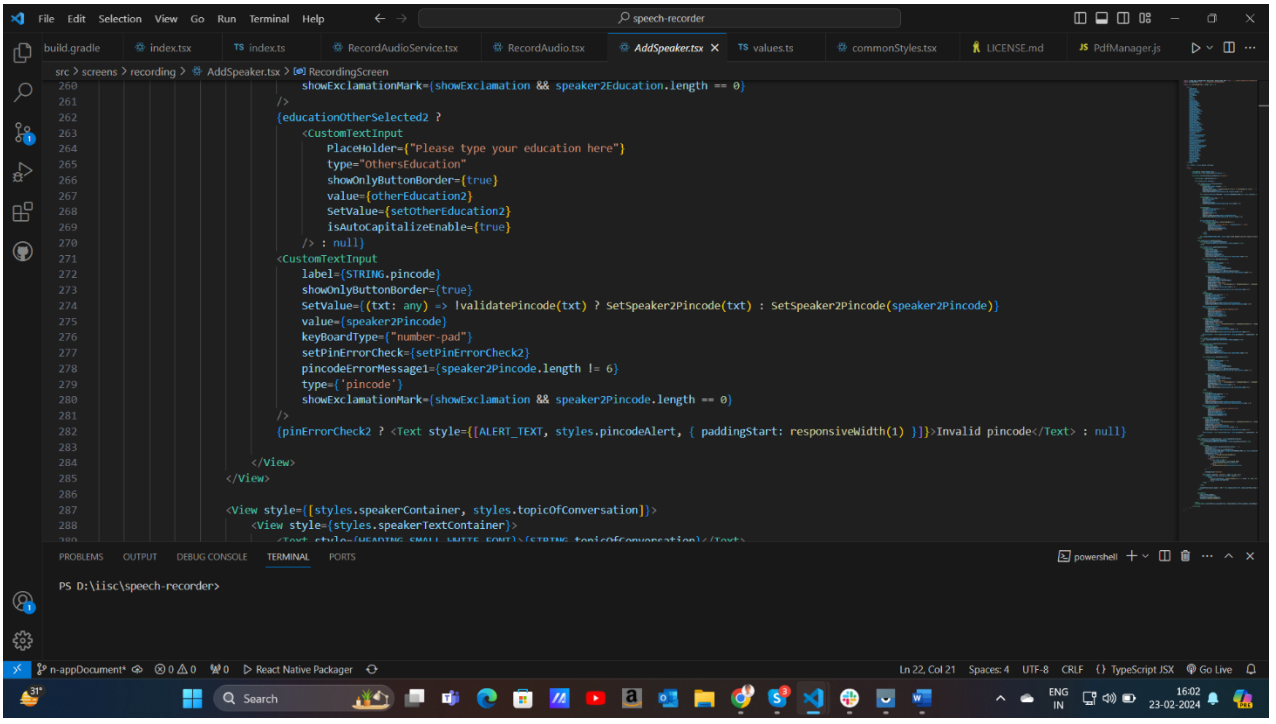


Fig. Filled Users Details screen code

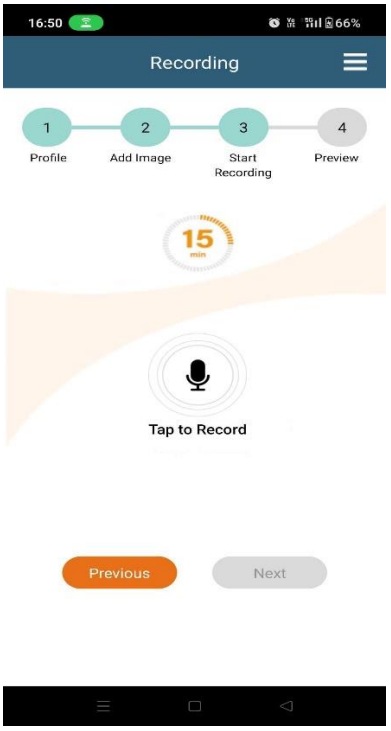


Fig. Start Recording Screen

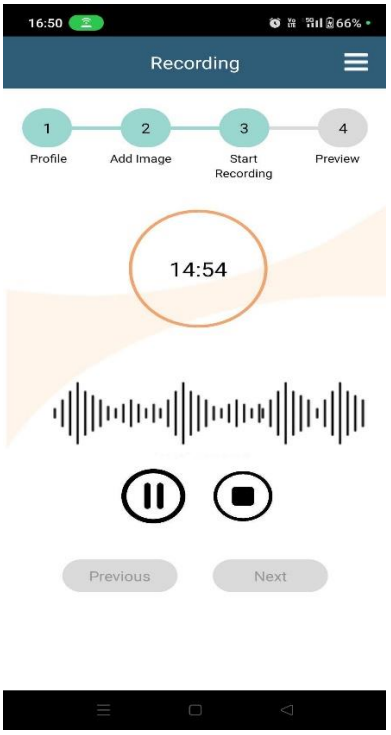


Fig. Recording Screen

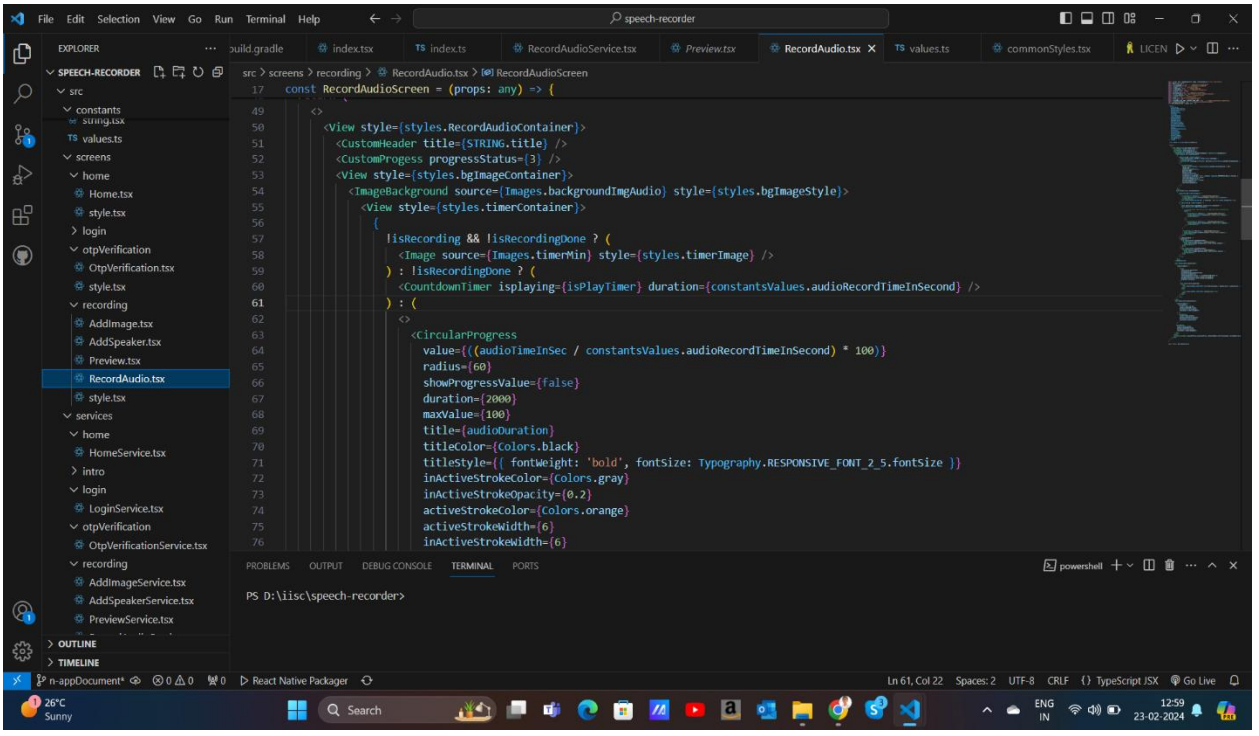


Fig. Start Recording screen code

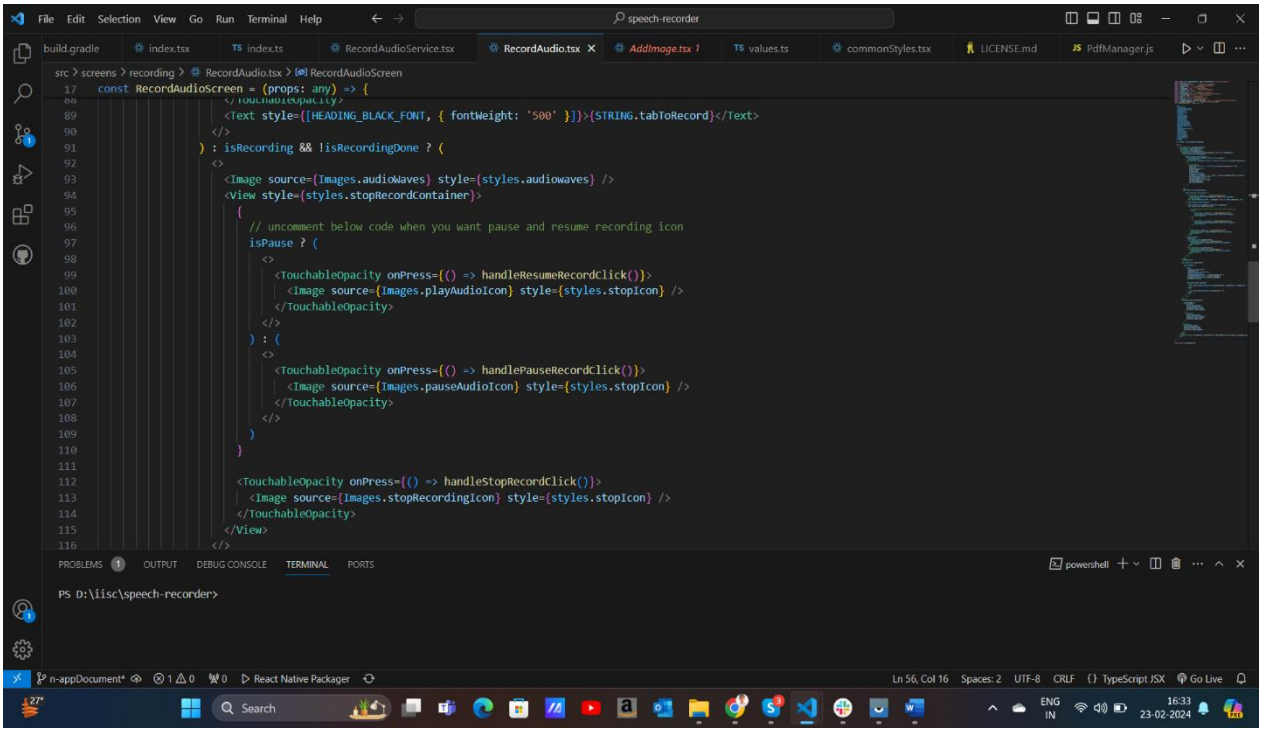


Fig. Recording Screen Code

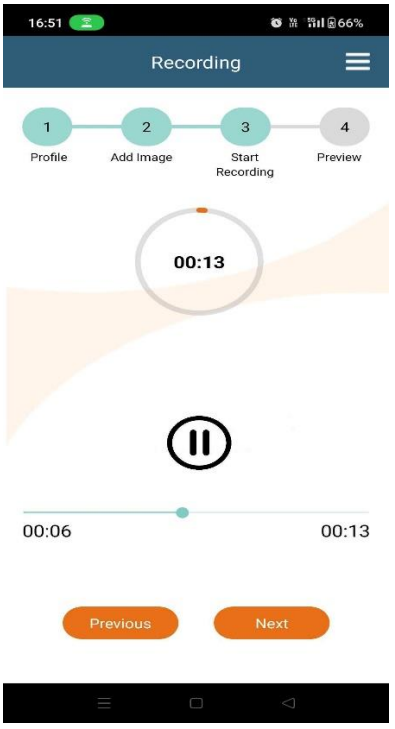


Fig. Play Audio Screen

Fig. Preview Screen

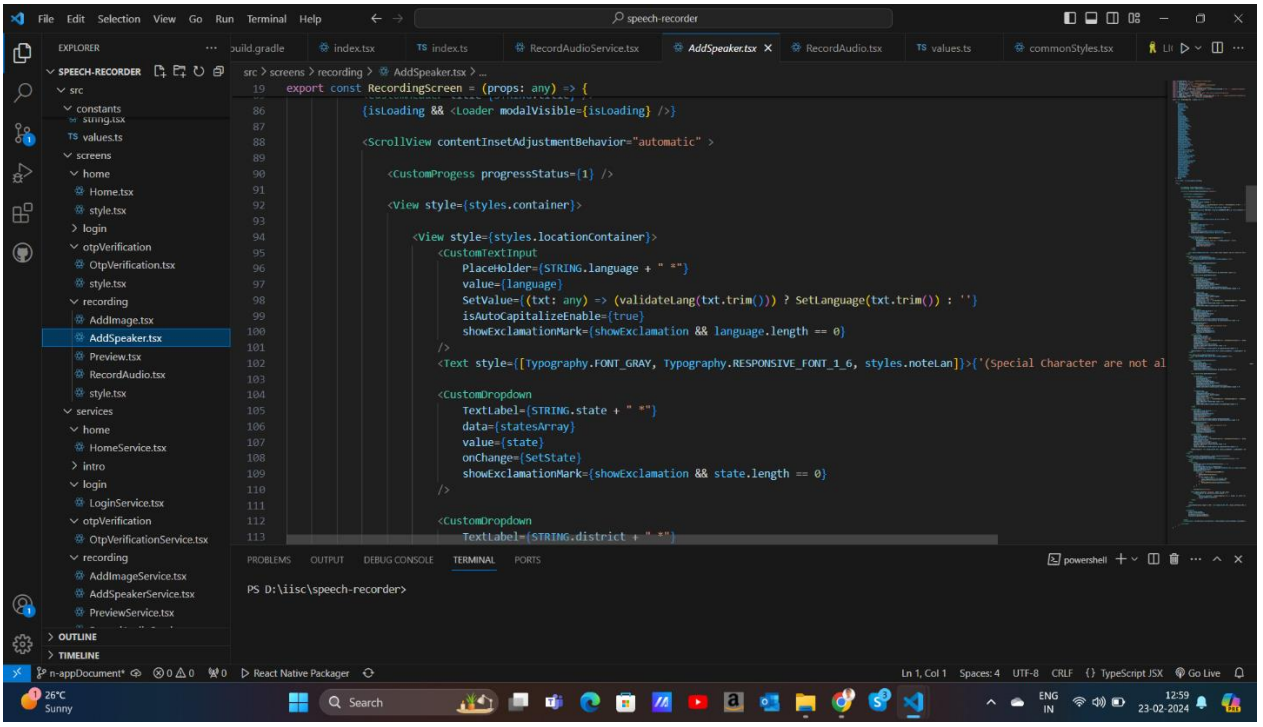


Fig. Play Audio Screen Code

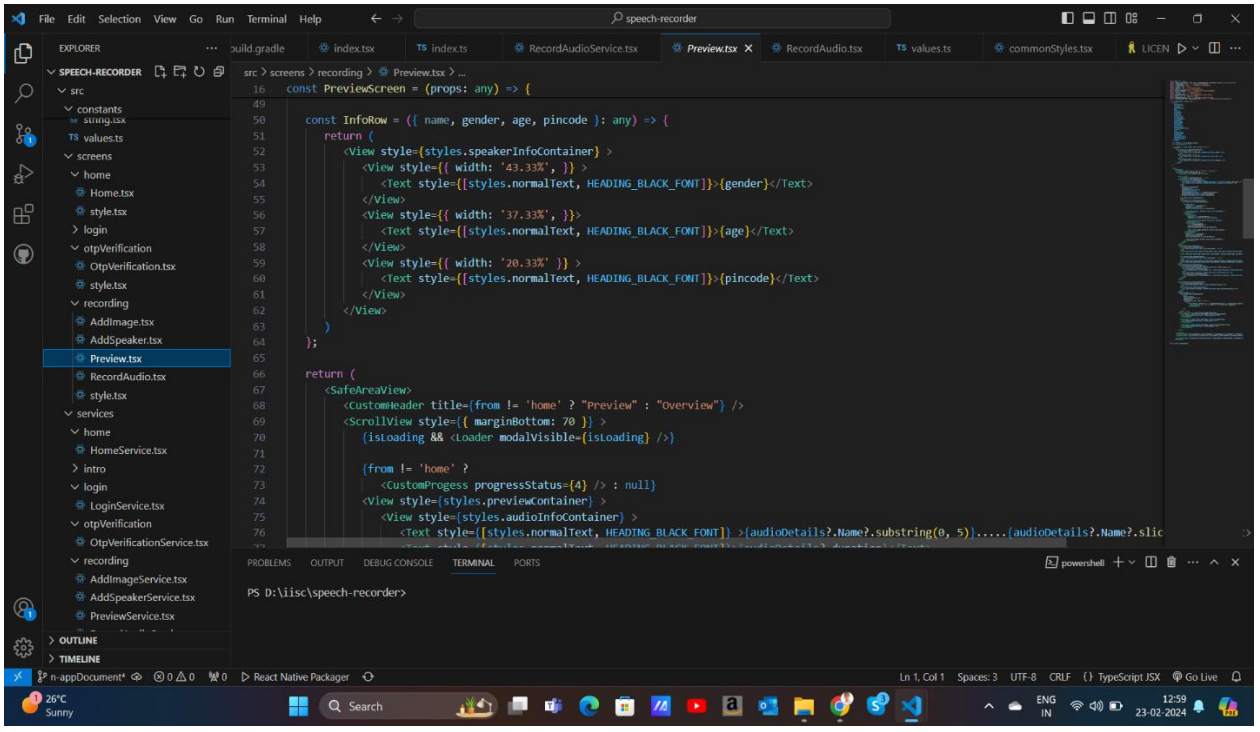


Fig. Preview Screen Code

Conclusion

7.1 Limitations

In our “My-Shop”, we are still improving and trying to add new functionalities. In “My-Shop” there is only “Wav Format” audio we can record. There are more options to add on “My-Shop” for better recording.

7.2 Future Enhancements

In the future, we will upgrade our Audio Recording app and add more Format support to do recording in “My-Shop”. In the future, we will add the functionality of drag & drop timeline of playing audio, Preview of the selected image and many more.

Bibliography

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